Applied Soft Computing 133 (2023) 109918

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Exploring deep feature-blending capabilities to assist glaucoma screening

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ARTICLE INFO

Article history:
Received 23 July 2022
Received in revised form 11 October 2022
Accepted 2 December 2022
Available online 9 December 2022

Keywords:
Deep learning
Optic disc and optic cup segmentation
Glaucoma diagnosis
Computer-assisted diagnosis
ESS-Net and FBSS-Net

ABSTRACT

Over the last three decades, computer vision has had a vital role in the healthcare sector by providing soft computing-based robust and intelligent diagnostic solutions. Glaucoma is a critical ophthalmic disease that can trigger irreversible loss of vision. The number of patients with glaucoma is increasing dramatically worldwide. Manual ophthalmic assessment of glaucoma detection is a tedious, error-prone, time-consuming, and subjective task. Therefore, computer-assisted automatic glaucoma diagnosis methods are required to strengthen existing diagnostic methods with their robust performance. Optic disc (OD) and optic cup (OC) segmentation have a key role in glaucoma detection. Accurate segmentation of the OD and OC provides valuable computational and clinical details that can substantially assist in the glaucoma screening process. Retinal fundus images have extensive variations in terms of size, shape, pixel intensity values, and background effects that make segmentation challenging. To mitigate these challenges, we developed two novel networks for accurate OD and OC segmentation. An efficient shallow segmentation network (ESS-Net) is the base network whereas a feature-blending-based shallow segmentation network (FBSS-Net) is the final network of this work. ESS-Net is a shallow architecture with a maximum-depth semantic preservation block for accurate segmentation, while FBSS-Net uses internal and external feature blending to improve overall segmentation performance.

To confirm their effectiveness, we evaluated both networks using four publicly available datasets; REFUGE, Drions-DB, Drishti-GS, and Rim-One-r3. The proposed methods exhibited excellent segmentation performance, requiring a small number of trainable parameters (3.02 million parameters).

1. Introduction

Technological advancements have revolutionized the global socioeconomic system. Specifically, artificial intelligence (AI) has ushered the world into a new automated era by introducing computer-assisted versatile applications [1]. AI has a vital role in social progress by providing intelligent, interactive, and robust diagnostic solutions [2]. AI and deep learning have also contributed significantly to healthcare by providing automated diagnosis, telehealth facilities, and intelligent biomedical assistance [3]. Globally, the incidence of retinal diseases is increasing at an alarming rate. Numerous ophthalmic diseases such as diabetic retinopathy, glaucoma, and macular edema can result in vision loss if not diagnosed at an early stage [4,5]. Deep learning-based methods can assist ophthalmologists in aiding the diagnosis of such critical ophthalmic diseases [6].

Glaucoma is the second most common neurodegenerative retinal disease and can cause irreversible vision loss [7]. Glaucoma development is attributed to increased intraocular pressure in the retina. Glaucoma cases are rapidly increasing every year, and according to one study, the total number of glaucoma cases in 2020 was estimated to be 80 million [8]. Glaucoma is more likely to develop in individuals over 60 years old. Considering the adverse effects of glaucoma, early detection is critical. Numerous methods are used to screen ophthalmic diseases; however, fundus imaging is considered a major tool for glaucoma detection owing to its rapid performance and low cost [9]. Optic cup (OC) and optic disc (OD) segmentation in fundus images is performed to aid the glaucoma screening process. The OD is a vertical oval-shaped area where retinal blood vessels enter the retina, whereas the OC is a cup-shaped bright area located at the center of the OD.

Glaucoma causes structural and morphological changes that result in cupping, which is referred to as OC enlargement. Similarly, the vertical cup-to-disc ratio (vCDR) is a key computational biomarker for the diagnosis of glaucoma [10]. The area-cup-to-disc-ratio (A-CDR), inferior superior nasal and temporal (ISNT) rule, and disc damage likelihood scale (DDLS) are indicators of glaucoma detection. A high segmentation performance is required for an accurate computational analysis of

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https://doi.org/10.1016/j.asoc.2022.109918

glaucoma assessment. Manual retinal assessment for glaucoma diagnosis is expensive, time-consuming, and requires specialized medical skills [11]. Therefore, computer-assisted diagnosis (CAD) is desirable for robust, cost-effective, and automated screening of glaucoma. CAD has ushered the diagnostic industry into a new era [12–14]. Specifically, deep learning-based methods are employed in many diagnostic applications because of their effective performance [15–17]. Extensive research is being conducted on CAD for automated diagnosis of glaucoma using neural networks [18]. However, existing methods suffered from several challenges and problems associated with OD and OC segmentation for glaucoma diagnoses. A few points described below enlighten the reason and gap in existing methods that trigger the development of new approaches. The OC boundary is typically indistinctive, which makes accurate segmentation challenging. State-of-the-art methods could not demonstrate an appealing performance, especially for OC. Subsequently, most methods are computationally expensive because they require extensive parameters for complete training of the network. Finally, images in different datasets are frequently significantly different in the acquisition process with variations in pixel intensity values, and backgrounds; therefore, maintaining high segmentation performance using multiple datasets is also challenging for existing methods. To address the aforementioned problems and challenges, we developed two novel shallow networks for accurate segmentation of OD and OC. Proposed methods are different from existing methods in many ways. According to the best of our knowledge, no existing methods used maximum-depth semantic preservation (MDSP) along with internal and external feature blending for glaucoma diagnosis or similar applications. In ESS-Net, the MDSP block is designed to preserve the semantic features that help in enhancing the accuracy of pixel-wise predictions. Internal and external feature blending-based structure is developed in FBSS-Net to further boost the segmentation performance. Contrary to most of the previous methods, the proposed methods use a shallow and computationally efficient architecture for high-efficiency training of the network. Architectures of ESS-Net and FBSS-Net do not contain a typical encoder–decoder structure in which the decoder is a mirror to that of the encoder. Instead, the decoder of both proposed networks uses only a single final convolutional layer for efficient training. As given in the subsequent section and Table 1, usually previous methods require preprocessing for achieving competitive results whereas the proposed methods do not need any processing. Both networks are developed from scratch and are not based on any other existing method.

The contributions of this study can be summarized as follows:

- Two architectures, an efficient shallow segmentation network (ESS-Net) and a feature-blending-based shallow segmentation network (FBSS-Net), were developed for accurate OD and OC segmentation with superior computational efficiency to assist the glaucoma screening and assessment process.
- ESS-Net is an independent segmentation network employed as the base network. ESS-Net has a MDSP block that preserves semantic information and improves prediction accuracy. FBSS-Net is the final network, and it uses internal and external feature blending to further improve segmentation performance.
- Both architectures were extensively evaluated on four publicly available datasets and exhibited excellent segmentation performance with superior computational efficiency (with 3.02 million trainable parameters). In addition, we have made our networks publicly available to other researchers for fair comparisons [19]

The remainder of this paper is organized as follows. Related research and the proposed method are discussed in Sections 2 and 3, respectively. The experimental results are presented in Section 4. Finally, the conclusions are presented in Section 5.

2. Related studies

The number of glaucoma patients worldwide is increasing. Therefore, fast, accurate, and automatic methods of glaucoma assessment are highly desirable. The contribution of deep learning-based methods to the automatic diagnosis of several diseases is significant [20–22]. Recently, many segmentation methods based on deep and handcrafted features have presented different approaches for glaucoma screening. For a discussion of the literature, previous studies are primarily divided into deep feature-based and handcrafted-based methods.

2.1. Handcrafted feature-based methods

Many studies have presented OD and OC segmentation approaches based on handcrafted features. In this type of segmentation, boundary problems are typically transformed into pixel-level classification problems [11]. Nevertheless, boundary extraction for the OD and OC is challenging because of variations in retinal fundus images and retinal blood vessel interventions [11]. Handcrafted feature-based segmentation methods can be further classified as follows.

2.1.1. Shape and morphology

In a previous study [23], OD segmentation was performed by eliminating peripapillary atrophy. Constraint Hough transform, edge detection, and atrophy detection aid in the elimination. The final results obtained using this method require post-processing, which can be indicated as the limitation of this method. Another method [24] employs an active shape model in combination with circular Hough transform and an edge detection mechanism for OD boundary extraction. This method achieves good performance in terms of average error; however, preprocessing is used to avoid retinal blood vessel intervention [24]. Similarly, some studies [25] used the morphological reconstruction of images as an effective step for OD boundary detection. The bright regions and green plane of the fundus images were morphologically reconstructed, and a convex hull was estimated for the final OD boundary extraction [25]. Although this method has a low computation time, it uses preprocessing and requires a preprocessing dataset [25]. In one study [26], mathematical morphology and principal component analysis were used for the automatic detection of the OD. This method was evaluated on five publicly available datasets, but the boundary extraction is limited to the OD.

2.1.2. Clustering and level-set

With this method [27], binary fitting energy is employed for OD segmentation, whereas a spatially weighted fuzzy c-means morphological and clustering approach is used for OC segmentation. This method has high sensitivity but requires preprocessing. Later, K-means clustering was used in [28] for the OD segmentation. In addition, fine OD contours were extracted using ellipse fitting and an active contour. This method also requires preprocessing to perform segmentation [28]. In another study [29], a hybrid approach combining clustering and level sets was employed for OD segmentation. Initially, clustering provided rough OD boundaries, whereas level-setting was applied to cover over- or under-segmentation cases [29].

Table 1
Methodological comparison between existing and proposed methods for the segmentation of OC and OD.

Type	Method	Approach/ Strength	Weakness
Handcrafted features-based methods	Shape and morphology [23–26]	These methods have less computational complexity.	Most of the methods require preprocessing and postprocessing
	Level-set and clustering [27–29]	Relatively possesses a high sensitivity in glaucoma screening. Can also cover over- and under-segmentation cases.	Variations in OD and OC affect performance. Most of the methods require preprocessing.
	Thresholding and texture [30,31]	Methods are generally capable to address variations in evaluation.	Methods require multi-stage preprocessing
Deep features-based methods	U-Net and modified U-Net [32–34]	U-Net is one of the famous and effective networks in medical image segmentation. Several methods modify U-Net to improve its performance with a smaller number of parameters requirement.	Preprocessing and a large number of parameters are required.
	Recurrent network [35,36]	Architectures with multi-path aid in obtaining semantic features. Recurrent networks also support in capturing high-level information.	These methods require preprocessing.
	Adversarial learning [37,38]	Adversarial learning enables the network to exhibit good performance with limited annotated data. Adversarial learning also helps in domain adaption.	Methods require preprocessing and/or postprocessing.
	Encoder–decoder structure [39,40]	Encoder–decoder structure-based methods provide input to the encoder and an output prediction mask is obtained from the decoder. Methods use different networks as encoders and decoders to achieve desired performance.	Pretrained weights are used. Relatively low segmentation performance for small-sized object (OC) prediction because of a very small final feature map size after downsampling.
	Transfer learning [41,42]	Transfer learning improves the segmentation performance in glaucoma screening. However, it triggers some computational trade-offs.	Pretrained weights, preprocessing, and transfer learning can be attributed as limitations of this method
	Attention mechanism [43,44]	Attention mechanism-based techniques add value in terms of improved performance and fast training of the network.	Cascading in case of false prediction will result in a transfer of false information and it can negatively affect the next stage performance. In addition, the network is evaluated using a single dataset.
	Fully convolutional network [45,46]	Fully convolutional networks improve overall performance for OC and OD segmentation. Feature re-use and different approaches for prioritizing the areas during the training process are also employed.	Preprocessing, a large number of trainable parameters, and a long training time.
	ESS-Net (proposed base network)	ESS-Net is an independent, efficient, and shallow network employed as the base network in this study for OC and OD segmentation. It has a maximum-depth semantic preservation (MDSP) block which preserves the semantic information and improves the prediction accuracy.	Segmentation performance can be relatively improved by feature blending.
	FBSS-Net (proposed final network)	FBSS-Net was the final network of this study, and it uses ESS-Net architecture as the base network. FBSS-Net employs the internal and external features blending for an improved segmentation with superior computational efficiency.	Proposed framework requires data augmentation for small datasets.

2.1.3. Thresholding and texture

In Ref. [30], a region-growing method based on adaptive thresholding was employed for OD segmentation. The method was evaluated on multiple datasets; therefore, adaptive thresholding was used to ensure high segmentation performance while addressing various images from different datasets [30]. Preprocessing to combine the green and red channels is required in this method [30]. Another study [31] utilized mainly the texture features for glaucoma screening in retinal fundus images. A correlation feature-selection scheme was employed to select these features. This method also requires preprocessing for the region of interest [31].

2.2. Deep feature-based methods

Deep feature-based segmentation exhibits robust performance in CAD. Unlike handcrafted features, deep feature-based methods do not require extensive pre- or post-processing [41]. In addition, deep learning-based methods can improve segmentation performance with a small dataset by using transfer learning [41].

Computer-assisted automated methods have superseded several traditional approaches in industry. Deep feature-based methods are further classified as follows:

2.2.1. U-Net and modified U-Net

U-Net [32] is one of the most famous and effective architectures for medical image segmentation. Several methods use the standard or modified U-Net for segmentation in different applications. In [33], U-Net was implemented on single-board computers, and segmentation for OD and OC was performed. The objective of this study was to confirm the effectiveness of U-Net in performing segmentation within an acceptable time using a single-board computer [33]. Similarly, another study [34] used a modified U-Net for glaucoma screening. In this method, the number of filters in the convolution layers was reduced to develop a lightweight architecture [34]. This method required preprocessing to obtain the reported results [34].

2.2.2. Recurrent networks

Recurrent architectures have also been employed for OD and OC segmentation. In [35], a U-Net-based multipath recurrent network was combined convolutional neural network (CNN) with a recurrent network for glaucoma screening [35]. The multipath architecture aided in obtaining semantic features from the encoder and decoder. This method requires polar transformation as a preprocessing step [35]. Similarly, another method [36] proposed a recurrent fully convolutional network (RFC-Net) for OC and OD segmentation. This method refers to a multi-scale input layer with edge information and high-level information captured by RFC-Net [36]. The limitations of this method are the evaluation using a single dataset and preprocessing requirements [36].

2.2.3. Adversarial learning

Adversarial learning has a vital role in CAD with limited annotated data. In one study [37], a generative adversarial network (GAN) was employed for OD segmentation from two datasets. The proposed network, called RetinaGAN, uses the U-Net architecture for the generator part, whereas different models are employed as discriminators [37]. Subsequently, in [38] Patch-based approach was employed for adversarial learning to overcome the domain shift problem. This method requires pre- and post-processing [38].

2.2.4. Encoder–decoder structure

Several deep learning-based networks use an encoder–decoder structure for the desired predictions. In [39], an encoder–decoder structure was used to perform multiple tasks. This method can generate pixel-level (segmentation) and image-level (classification) predictions [39]. The relatively low performance for OC segmentation can be considered the weakness of this framework [39]. Another study [40] developed a simple method that employs residual network-34 (ResNet-34) as the encoder, while the decoder part is based on U-Net. This method requires less training time; however, it uses pretrained weights for initialization [40].

2.2.5. Transfer learning

Transfer learning is used in many CNNs to improve network performance. In [42], DeepLabv3+ with multiple CNNs in the encoder was used for the segmentation of the OD. Subsequently, classification was performed using pretrained networks with the aid of transfer learning [42]. Preprocessing, pretrained weights, and transfer learning requirements can be considered the limitations of this method [42]. One method [41] developed a particle swarm optimization algorithm-based network for retinal fundus image segmentation. Considering the segmentation capabilities of a mask R-CNN, it uses transfer learning with an optimization algorithm for improved segmentation [41]. This method is limited to OD segmentation [41].

2.2.6. Attention mechanism

Attention mechanism-based techniques add value to the efficient training of CNNs. In [43], the attention mechanism was used for fast training, resulting in the early convergence of the network. A cascaded framework using anatomical knowledge was employed for the accurate segmentation of the OD and OC [43]. Cascading in the case of false prediction will result in the transfer of false information, which can negatively affect the next-stage performance [43]. Similarly, in [44], a self-attention method was used in a deep CNN to compute the CDR for glaucoma diagnosis. This method proposed an encoder–decoder structure in which self-attention was developed in the encoder section to improve performance [44]. Network evaluation using a single dataset can be considered a limitation of this method [44].

2.2.7. Fully convolutional network

Fully connected layer-based segmentation architectures are also commonly used in methods that provide CAD. In [45], a fully convolutional network with DenseNet was proposed for OC and OD segmentation. This method also introduces feature reuse, which lowers the number of required trainable parameters [45]. The preprocessing requirements and relatively long training time can be considered the limitations of this method [45]. Similarly, another study [46] used a fully convolutional network to assist in the glaucoma screening process. This study also highlighted techniques that can be employed to prioritize different regions during training [46]. This method requires a large number of trainable parameters [46].

A summary of related studies, discussing their strengths/approaches and weakness/limitations is provided in Table 1.

3. Proposed methods

3.1. Overview of the proposed methods

Using retinal fundus images, OC and OD segmentation is performed to aid the glaucoma screening process. An overview of the proposed method is shown in Fig. 1. Considering the memory and time limitations, the original images from the training split were initially resized for fast training of the network. CNNs require extensive annotated data for optimal learning. Data augmentation was employed to create sufficient data for the network training. The proposed networks were trained using augmented training images for given classes (OD/OC), whereas evaluation was performed on unseen images from the testing split, and networks provided segmented OD/OC as output. Resizing of segmented output back to the original size is performed to fairly compare with the ground truth images. Finally, in the evaluation stage, the segmented output was compared with the ground-truth image, and both quantitative and qualitative results were used for performance analysis.

3.2. Segmentation of OC and OD using ESS-Net

ESS-Net is an independent complete network for the segmentation of the OD and OC. In this study, ESS-Net was employed as the base network. The ESS-Net architecture is shown in Fig. 2. A retinal image with OD/OC is provided to the first layer of the encoder as the input. The ESS-Net uses seven convolution layers with one grouped convolution layer in the encoder. ESS-Net is an efficient network because of the efficient decoder used in its architecture. Unlike the conventional decoders in CNNs, ESS-Net does not have an ordered series of convolution layers in its decoder. However, it uses only a single 1×1 pointwise convolution layer, which is used for final feature extraction along with channel assignment according to the number of classes. The

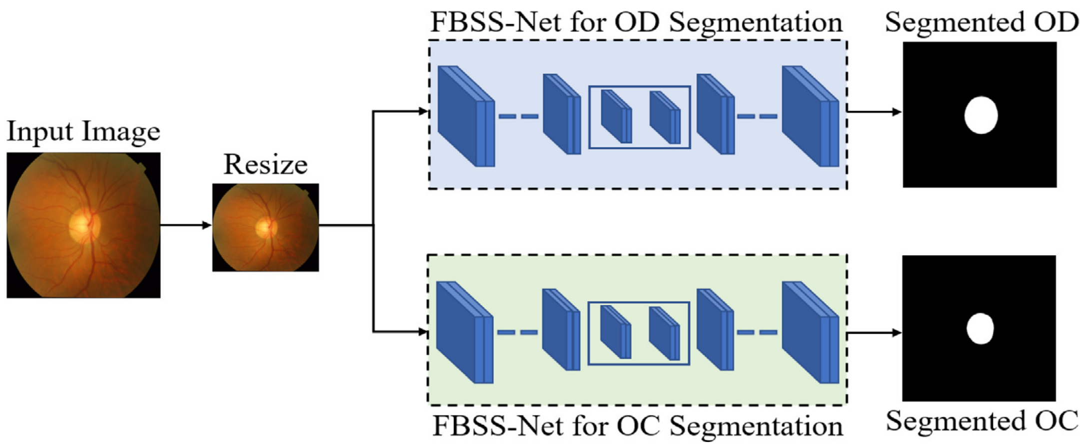


Fig. 1. Overview of the proposed method.

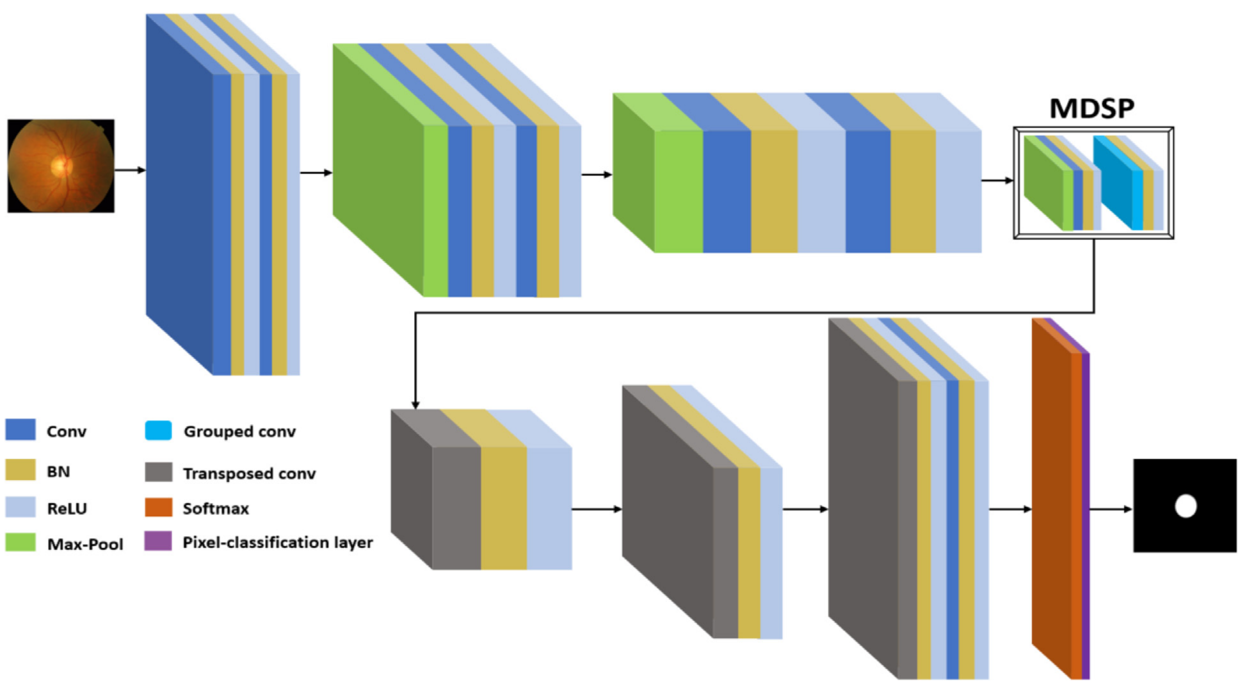


Fig. 2. Proposed ESS-Net architecture.

reason for not using multiple convolution layers in the decoder is to avoid complexity, spatial loss, and reduction of trainable parameters.

Many studies [47] have reported that deeper layers in any CNN contain rich semantic details. Therefore, ESS-Net adds an MDSP block at its maximum depth, where the spatial dimension is minimum whereas the number of channels is maximum (512). The MDSP block has two sets of convolutions: batch normalization (BN) and rectified linear unit (ReLU) layers with one max-pooling (max-pooling) layer. One of the convolution layers in the MDSP is selected as a grouped convolution because of its lower parameter requirement. The combination of convolution layers with ReLU and BN helps in preserving the rich semantics and consequently results in enhanced performance. The output from the MDSP block is upsampled using three transposed convolutional layers in the decoder. A prediction mask is generated for the desired class predictions using a pixel classification layer. Four public datasets are tested for the evaluation of ESS-Net. Nevertheless, ESS-Net showed promising performance for all datasets.

3.3. Segmentation of OD and OC using FBSS-Net

FBSS-Net, an enhanced version of ESS-Net, was developed for enhancing the accuracies of ESS-Net. In CNNs, features undergo spatial information loss during the downsampling and upsampling processes. As shown in Fig. 3, feature blending was employed to overcome spatial information loss with improved learning. Four internal feature blending points (IFB1–4) were included for each feature map size level in the encoder. The IFB permeates spatial information loss and strengthens spatial features. In a CNN, spatial information sharing between the encoder and decoder is crucial for improved learning. Three external feature blending points (EFB1–3) were added to the decoder for feature blending between the encoder and decoder at multiple levels. Features blending between the encoder and decoder through EFB significantly enhances the learning of the network. FBSS-Net was also tested using four public retinal fundus image databases and exhibited state-of-the-art segmentation performance.

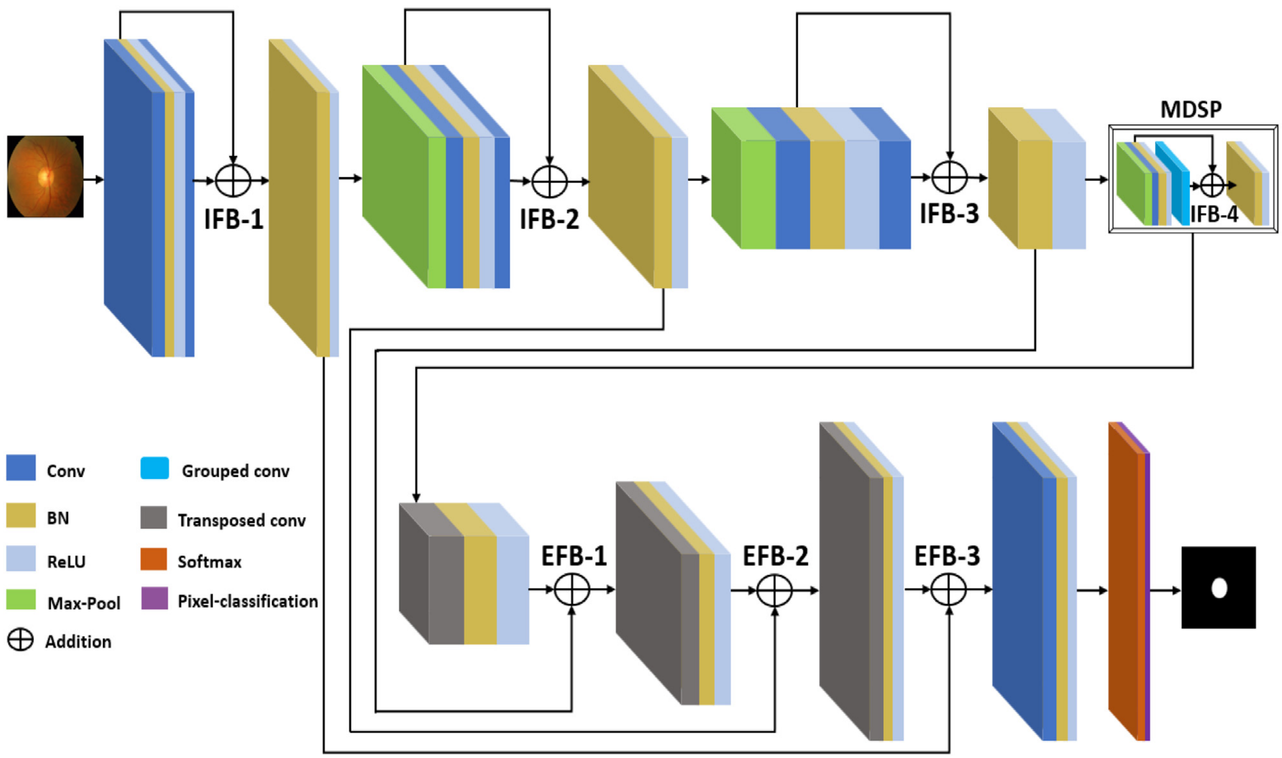


Fig. 3. Proposed FBSS-Net architecture.

$$FB_f = E_f + \Delta IB_{sf} \tag{3}$$

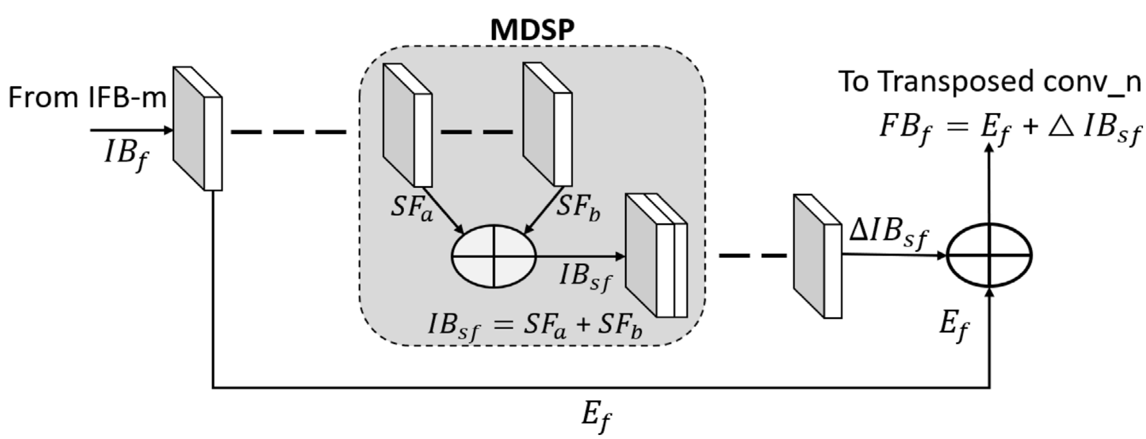


Fig. 4. Schematic diagram showing internal and external feature blending.

The schematic diagram shown in Fig. 4 shows the internal and external features of the blending process. In the MDSP block, internal semantic features (SF_a) from one layer are blended with the semantic features (SF_b) of the second layer, providing internally blended semantic features (IB_{sf}), as follows:

$$IB_{sf} = SF_a + SF_b \tag{1}$$

This internally blended semantic feature is processed through network layers, and the change produced is denoted as "Δ" as follows:

$$\Delta IB_{sf} = \Delta(SF_a + SF_b) \tag{2}$$

Similarly, a spatial feature from the last IFB stage is shown with IB_f. The processed feature in the encoder to be blended in the decoder is called the spatial external feature (E_f). On the right side of the schematic diagram, E_f is blended with an internal blended semantic feature, providing the final blended feature (FB_f) as the output. FB_f has a valuable impact on the learning process because it combines external features with internally blended features.

$$FB_f = E_f + \Delta IB_{sf} \tag{3}$$

Feature blending mechanism and MDSP in the proposed FBSS-Net are different from existing methods. Considering SLSR-Net [48], it only uses the external residual connection. SLSR-Net fuses the input feature map with a processed feature through a separable convolution link (SCL).

$$S_{EW} = G_{SC} + I_{FM} \tag{4}$$

In the above expression, I_{FM} represent input feature map whereas G_{SC} refers to the processed feature by SCL. The proposed FBSS-Net uses both external and internal feature blending simultaneously.

As shown in Eq. (1), internal semantic features (SF_a) are blended with the semantic features (SF_b) of the preceding layer and delivers internally blended semantic features (IB_{sf}) at the output. Subsequently, Spatial external features (E_f) are blended with the processed features of IB_{sf} as shown in the final Eq. (3) of FBSS-Net. The proposed model achieved a better performance because of the internal and external feature blending along with the MDSP mechanism. In FBSS-Net, internal feature blending is performed four times whereas external feature blending is carried out three times. Internal feature blending provides feature empowerment after intervals. These internally blended features are further blended with external features at different stages of the network, resulting in diverse learning of the network.

The FBSS-Net layer configuration is provided in Table 2. Details regarding the layers' names, sizes, number of filters, output features, and number of required parameters are included in this table. After acquiring the image features using the image input layer, the first convolutional layer (Conv-1) applies 64 filters and after activations it provides 600 × 600 × 64 as the output features. Similarly, featuremap size is initially downsampled using pooling operations, and the number of filters increases with reduced featuremap size. Later, Featuremap size is upsampled using transposed convolutional layers, and the number of applied filters decreases with upsampled featuremap size. The number of required parameters is directly associated with the computational efficiency of the network and is provided in the last column of Table 2. In Table 2, only those layers are mainly included which require trainable parameters. It is notable, the final featuremap size (75 × 75 × 512) is deliberately kept sufficiently large so that network can maintain its performance even for minor image details.

4. Experimental results

4.1. Experimental details and environment

In this study, experiments are performed on four public databases: REFUGE[1] [49], digital retinal images for optic nerve segmentation database (Drions-DB)[2] [50], reference image database (Rim-One-r3)[3] [51], and Drishti-GS[4] [52]. REFUGE is one of the latest and most challenging datasets launched in 2018. The REFUGE dataset has 400 testing, 400 training, and 400 validation images with a size of 2124 × 2056 pixels. As shown in Fig. 5, REFUGE contains joint annotations for the OD and OC. In the ground-truth image, the OD and OC regions are represented by gray and black colors, respectively, whereas the background is white.

The Drions-DB dataset from Miguel Servet Hospital (Spain) has 110 images with OD annotations. The size of all images in this dataset is 600 × 400 pixels, and the same image size was used in the experiments. Drions-DB example images are given in Fig. 6. The white and black regions in the ground truth image represent OD and background regions, respectively.

Rim-One-r3 is one of the most challenging databases for OD and OC segmentation. It has a total of 159 images and was launched by the MIAG group, University of La Laguna, Spain. A sample image from the Rim-One-r3 dataset is shown in row 2 of Fig. 7. Aravind Eye Hospital introduced Drishti-GS dataset in 2014 and it contains 101 images. It has training and testing splits comprising 50 and 51 images, respectively. A sample image from the Drishti-GS dataset is shown in row 1 of Fig. 7. The OD and OC regions are shown in white in Figs. 7(b) and 7(c), respectively.

The testing and training of both networks were performed using a desktop computer with an Intel® Core™ i7 CPU950@3.7 GHz processor, 32 GB of random access memory (RAM), and an NVIDIA GeForce GTX 1070 graphics processing unit (GPU) with 8 GB graphics memory [53]. The MATLAB 2020b [54] framework was used for network development, training, and evaluation.

Segmentation requires annotated data for the training and evaluation of the networks. Limited annotated data is a major limitation in medical image segmentation. To overcome this problem, data augmentation was used to artificially create data for training purposes. In our study, geometric and arithmetic operations were performed for augmentation. This includes flipping, translation, cropping, and a combination of these operations to create additional data.

4.2. Training of the proposed networks

Considering the training time and memory constraints, the training images were resized. In our experiments, REFUGE, Drions-DB, Drishti-GS, and Rim-One-r3 datasets were resized to 600 ×600, 600×400, 500× 500, and 600 × 400 pixels, respectively. Training and testing splits were predefined for the REFUGE and Drishti-GS datasets. REFUGE has a total of 1200 images and its data is divided into three equal parts of 400 images for training, validation, and testing by the dataset provider. Similarly, Drishti-GS training and testing images are 50 and 51, respectively. In addition, 10% of the Drishti-GS training images are randomly selected as validation. For Rim-One-r3, the same data split provided in [38] was followed and randomly 99 images are selected for training/validation and 60 images for testing purposes. Similarly, the same data split as provided by [37] was followed for Drions-DB and used 100 images for training/validation and 10 images for testing. In our work, no preprocessing is involved for all four datasets. The Adam optimizer was employed for optimization because of its fast convergence, data-handling capabilities, and adaptive learning [55]. Both ESS-Net and FBSS-Net were trained independently for all four datasets. Early stopping and augmentation were used to avoid overfitting. In addition, Tversky loss was employed as the loss function for efficient network training for all four datasets (details are presented in the next subsection). The training accuracy and loss plots of the Rim-One-r3 dataset shown in Fig. 8 indicate efficient training with minimal loss. As shown in Fig. 8, the training of OD used 17 epochs with 8381 iterations whereas OC took 13 epochs with 8229 iterations. Parameter details for both OD and OC are the same as follows except for the minibatch size which is 8 and 7 for OD and OC, respectively. Initial learning rate: 0.001; Squared gradient decay factor: 0.95; L2 regularization: 0.0005; Epsilon: 0.000001.

The Tversky [56] loss function was employed in the training of both networks. In retinal fundus images, the OD and OC area is relatively small compared to the background. This scenario introduces a class imbalance problem. Subsequently, OC segmentation becomes challenging because of its non-distinctive boundaries; therefore, effective training of the network is required to achieve high segmentation performance. To overcome these challenges simultaneously, we used Tversky loss as the objective function, considering its capabilities to overcome class imbalance problems while delivering superior segmentation performance. Tversky loss guides the network for optimized performance, ensuring higher training accuracies with minimal loss. The trade-off between false positives and negatives can also be settled using the training

1 https://ai.baidu.com/broad/download?dataset=gon
2 http://www.ia.uned.es/~ejcarmona/DRIONS-DB.html
3 http://medimrg.webs.ull.es/rim-one-release-3-is-finally-here/
4 http://cvit.iiit.ac.in/projects/mip/drishti-gs/mip-dataset2/Home.php

Table 2
FBSS-Net layers configuration details and required number of trainable parameters (Configuration is based on REFUGE dataset with the size of 600 × 600 pixels).

Layer Name	Size	No. of filters	Output features (width × height × number of channels)	Parameters
Conv-1 + ReLU-1	3 × 3 × 3	64	600 × 600 × 64	1,792
BN-1	–	–		128
Conv-2 + ReLU-2	3 × 3 × 64	64		36,928
BN-2	–	–		128
Max-pool-1	3 × 3	–	300 × 300 × 64	–
Conv-3 + ReLU-3	3 × 3 × 64	128	300 × 300 × 128	73,856
BN-3	–	–		256
Conv-4 + ReLU-4	3 × 3 × 128	128		147,584
BN-4	–	–		256
Max-pool-2	3 × 3	–	150 × 150 × 128	–
Conv-5 + ReLU-5	3 × 3 × 128	256	150 × 150 × 256	295,168
BN-5	–	–		512
Conv-6 + ReLU-6	3 × 3 × 256	256		590,080
BN-6	–	–		512
Max-pool-3	3 × 3	–	75 × 75 × 256	–
Conv-7 + ReLU-7	3 × 3 × 256	512	75 × 75 × 512	1,180,160
BN-7	–	–		1,024
Conv-8 + ReLU-8	3 × 3 × 256	512		5,120
BN-8	–	–		1,024
Tra-conv-1	2 × 2	256	150 × 150 × 256	524,544
ReLU-9+ BN-9	–	–	150 × 150 × 256	512
Tra-conv-2	2 × 2	128	300 × 300 × 128	131,200
ReLU-10+ BN-10	–	–	300 × 300 × 128	256
Tra-conv-3	2 × 2	128	600 × 600 × 64	32,832
ReLU-11+ BN-11	–	–	600 × 600 × 64	128
Conv-9 + ReLU-12	3 × 3 × 256	3	600 × 600 × 3	1731
BN-12	–	–		6
Total number of trainable parameters				3,025,737

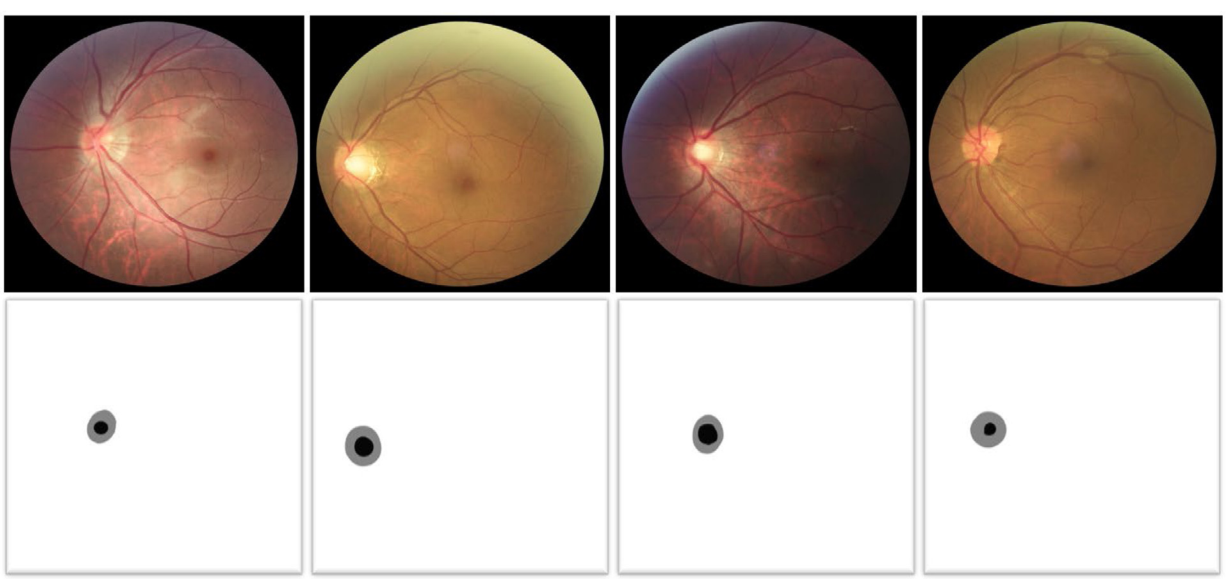


Fig. 5. REFUGE dataset sample images with ground truth.

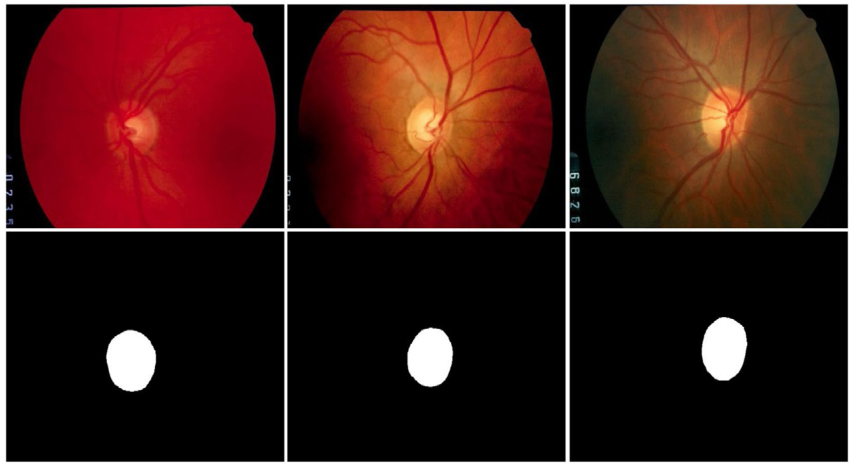
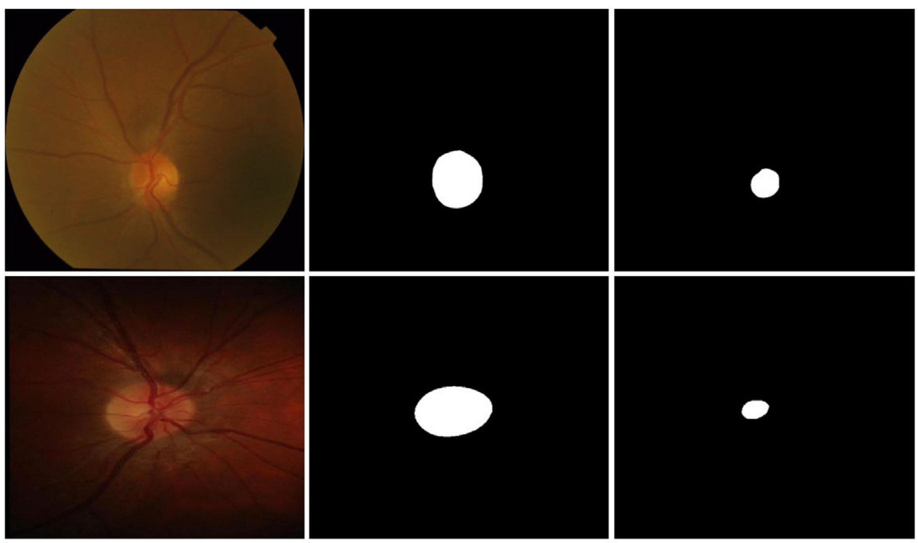


Fig. 6. Drions-DB sample images with ground truth.



(a) (b) (c)

Fig. 7. Example images of row 1: Drishti-GS; row 2: Rim-One-r3. (a) Original image (b) ground truth image with OD (c) ground truth image with the OC.

components of Tversky loss [57]. The mathematical expression for the Tversky loss is as follows:

$$T_{Loss} = \frac{\sum_{i=1}^{N} P_b^i G_b^i + \omega}{\sum_{i=1}^{N} P_b^i G_b^i + \omega + \alpha \sum_{i=1}^{N} P_{nb}^i G_b^i + \beta \sum_{i=1}^{N} P_b^i G_{nb}^i + \omega} \tag{5}$$

In this equation, α and β are the training components that can be initially set to optimize the training given that ($\alpha + \beta = 1$). In our case, an initial ratio of $\alpha{:}\beta = 0.7{:}0.3$ was adopted to complete the training process. G_b^i and P_b^i represent the pixels of the desired classes (OD/OC) and undesired classes in the ground truth, respectively.

4.3. Proposed method testing

ESS-Net and FBSS-Net were evaluated using four publicly available datasets to confirm their effectiveness. The original image was initially resized using nearest-neighbor interpolation. A trained network was applied to the input image, and a prediction mask was generated for the prediction of the OD and OC classes. To fairly compare with ground truth image, the prediction mask is resized back to its original size using nearest-neighbor interpolation. The ground truth image and generated mask are compared, and the evaluation was conducted based on performance measures. The same process is followed for testing all datasets. Independently trained networks for each dataset are applied for the testing of the networks. Testing is solely performed on unseen testing splits of respective datasets. Both ESS-Net and FBSS-Net can perform multiclass pixel-wise segmentation. The same testing criteria and evaluation measures are used to evaluate both ESS-Net and FBSS-Net for a fair comparison. Details of the evaluation measures are as follows.

For the evaluation of the proposed methods, well-known segmentation evaluation measures were employed [58]. These included accuracy (AC), sensitivity (SE), specificity (SP), Dice similarity coefficient index (DI), and Jaccard index (JC). Among these performance evaluation measures, DI is considered a major evaluation tool, particularly in medical image segmentation applications. Pixel-wise predictions were compared with the ground truth, and numerical results were calculated based on true negative (*tn*), true positive (*tp*), false negative (*fn*), and false

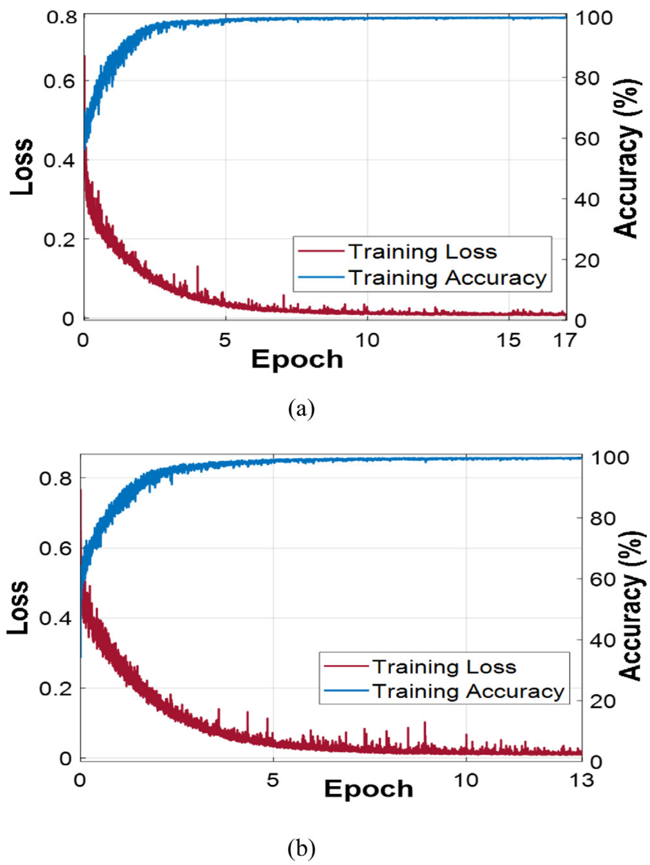


Fig. 8. FBSS-Net training accuracy and loss curves for the (a) OD and (b) OC.

positive (fp) computations. The mathematical expressions for the performance measures are as follows:

$$AC = \frac{tp + tn}{tp + fn + fp + tn} \tag{6}$$

$$SE = \frac{tp}{tp + fn} \tag{7}$$

$$SP = \frac{tn}{tn + fp} \tag{8}$$

$$DI = \frac{2tp}{2tp + fp + fn} \tag{9}$$

$$JC = \frac{tp}{tp + fp + fn} \tag{10}$$

4.3.1. Comparison of ESS-Net and FBSS-Net numerical results for segmenting OC and OD segmentation using all datasets of this work (ablation studies)

Proposed methods, ESS-Net and FBSS-Net, are evaluated using four datasets. Both networks delivered an excellent performance for OD and OC segmentation. However, FBSS-Net demonstrated better performance compared with ESS-Net because of its architectural effectiveness. The numerical results comparison between ESS-Net and FBSS-Net is shown in Table 3. Results of ablation studies present a superior performance from FBSS-Net for most of the evaluation measures. In semantic segmentation, DI and JC are commonly considered the main evaluation measures to assess segmentation performance. It is evident from Table 3, FBSS-Net results for DI and JC are better compared with ESS-Net for all considered datasets. All four databases of this study contain extensive inter-dataset and intra-dataset variations that make their accurate segmentation challenging. Nevertheless, FBSS-Net which is the final network of this study optimally overcome these challenges and manages to provide better segmentation performance. Internal and external feature blending, along with the MDSP role is possibly the reason for the performance difference. Feature blending mechanism helps in improving prediction accuracy, particularly for the boundaries of the OC and OD.

4.3.2. Comparison of ESS-Net and FBSS-Net visual results for segmenting OC and OD with REFUGE dataset (ablation studies)

The visual results for the good segmentation of OC and OD using both proposed networks are presented in Fig. 9. The original input images, corresponding ground truth images, segmented output images using ESS-Net, and segmented output images using FBSS-Net are shown in columns 1, 2, 3, and 4 of Fig. 9, respectively. A close visual analysis of the results indicated better segmentation performance using FBSS-Net compared with ESS-Net. REFUGE is one of the most challenging fundus image datasets for OC and OD segmentation. REFUGE has a variety of images with different OC and OD shapes and different pixel intensity values. Specifically, most images have indistinctive boundaries in the OC region. These problems make segmentation challenging. However, FBSS-Net mitigates these variations and provides better performance compared with ESS-Net. This performance improvement is because of the internal and external features blending at different stages of the network. In addition, visual results clearly show a high-class imbalance, nevertheless, proposed networks minimize this problem using Tversky loss and manage to deliver remarkable results. Some of the poorly segmented images of the sample are shown in Fig. 10. The poor segmentation of the given images can be attributed to the indistinctive boundaries of the OC and different pixel intensity values of the OD. Despite these challenges, FBSS-Net exhibited better segmentation than ESS-Net.

4.3.3. Comparing the proposed method on the REFUGE database with those of state-of-the-art methods

Performances of ESS-Net and FBSS-Net were further compared with other methods. The comparative numerical results presented in Table 4 confirm that FBSS-Net outperformed existing methods because of its semantic preservation and feature blending mechanism.

4.3.4. Comparison of ESS-Net and FBSS-Net visual results for OC and OD segmentation on the Rim-One-r3 dataset (ablation studies)

The superior segmentation performance using FBSS-Net compared with ESS-Net is depicted in Figs. 11 and 12. In Figs. 11 and 12, good segmentation visual results are shown in the first two rows (rows 1 and 2), whereas relatively poor segmentation visual results are shown in the last row (row 3). The relatively poor OD segmentation result in Fig. 11 (row 3) was likely because of the variation in pixel intensity values. Subsequently, the poor segmentation result shown in Fig. 12 (row 3) can be attributed to the small size of OC with indistinctive boundaries. Nonetheless, FBSS-Net delivers better segmentation performance for all cases. Rim-One-r3 is also a challenging dataset because of the extensive variations. Rim-One-r3 has variations in terms of size, shape, and pixel intensity values for OD and OC. Internal and external feature blending in FBSS-Net enables the network to better learn these variations and results in improved performance.

4.3.5. Comparing the proposed method numerical results on Rim-One-r3 database with those of state-of-the-art methods

A comparison between proposed methods and state-of-the-art methods is presented in Table 5 for a comparative assessment of

Table 3
Numerical performance comparison between ESS-Net and FBSS-Net using REFUGE, Rim-One-r3, Drions-DB, and Drishti-GS datasets (unit: %).

Datasets	Proposed methods	OC					OD				
		AC	SE	SP	DI	JC	AC	SE	SP	DI	JC
REFUGE	ESS-Net	99.92	85.91	99.98	89.20	81.60	99.88	97.98	99.91	96.26	92.91
	FBSS-Net	99.92	91.95	99.96	90.70	83.53	99.91	97.41	99.95	97.08	94.39
Rim-One-r3	ESS-Net	99.59	89.70	99.73	85.63	75.43	99.69	97.48	99.74	96.48	93.23
	FBSS-Net	99.62	89.63	99.76	86.38	76.61	99.74	97.04	99.86	97.02	94.23
Drions-DB	ESS-Net	–	–	–	–	–	99.76	94.45	99.95	96.44	93.17
	FBSS-Net	–	–	–	–	–	99.82	97.41	99.91	97.40	94.95
Drishti-GS	ESS-Net	99.71	92.69	99.85	90.91	84.48	99.85	96.45	99.97	97.74	95.64
	FBSS-Net	99.72	94.0	99.84	91.75	85.61	99.87	97.87	99.93	98.04	96.19

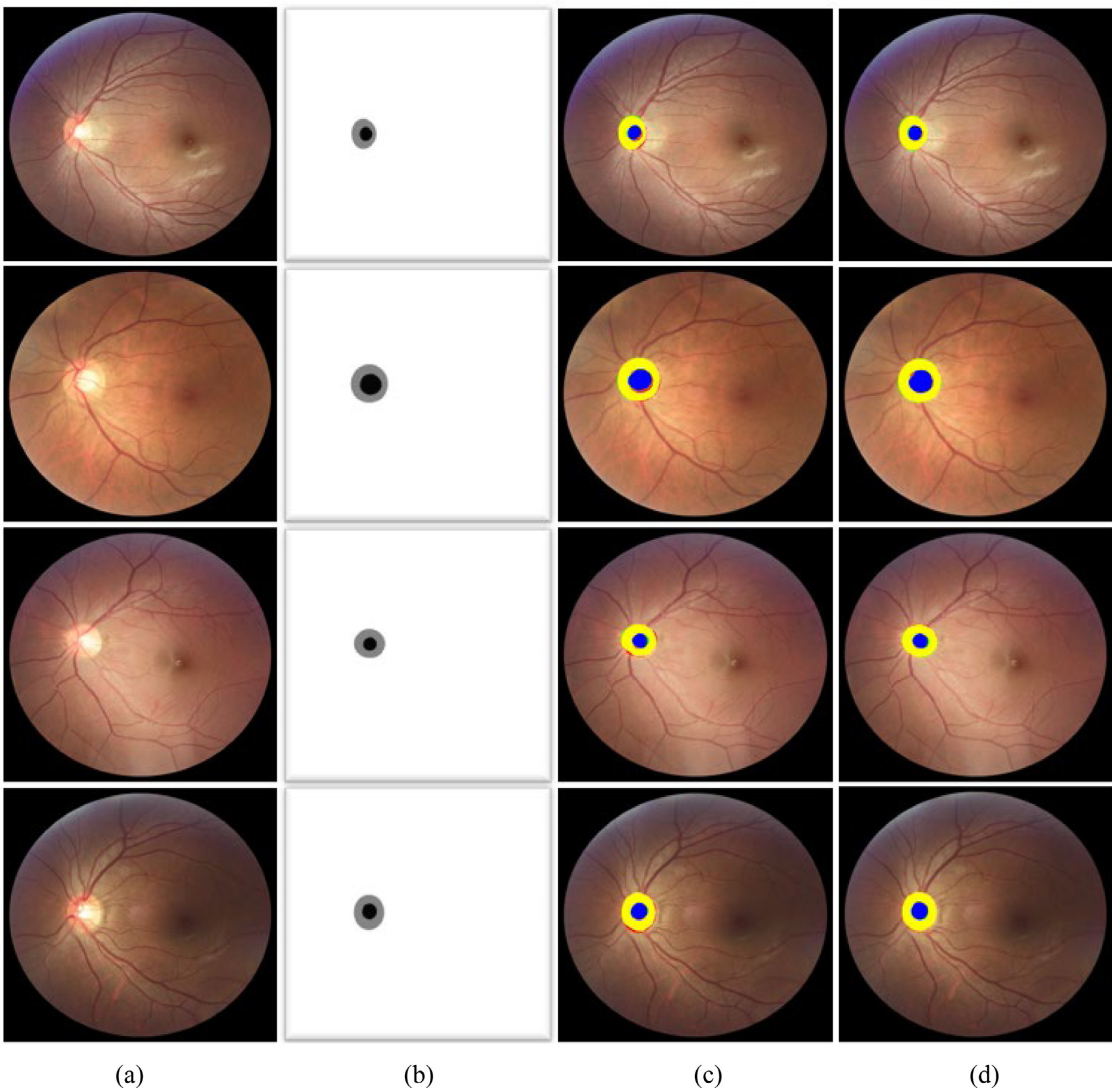
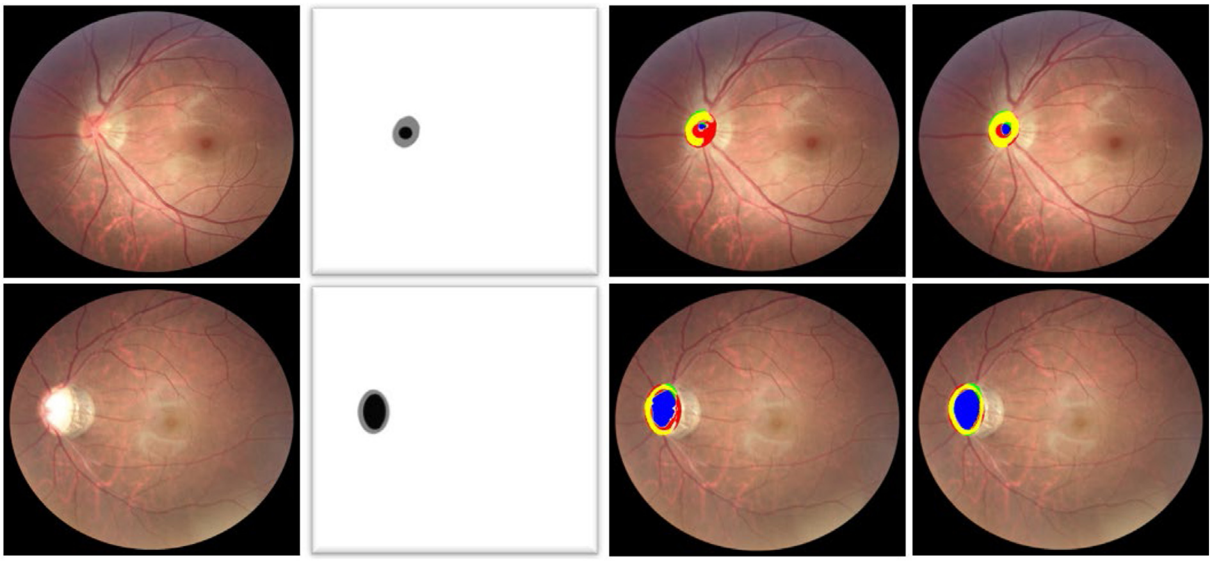


Fig. 9. Good OC and OD segmentation visual results on the REFUGE database, attained using proposed ESS-Net and FBSS-Net. (a) Input image, (b) corresponding ground truth image, and segmented output images acquired by applying (c) ESS-Net and (d) FBSS-Net (tp pixels for OD and OC are indicated in yellow and blue, respectively. Similarly, red and green indicate fn and fp pixels, respectively).

proposed methods. FBSS-Net outperformed the existing methods in both OC and OD segmentation for all performance measures.

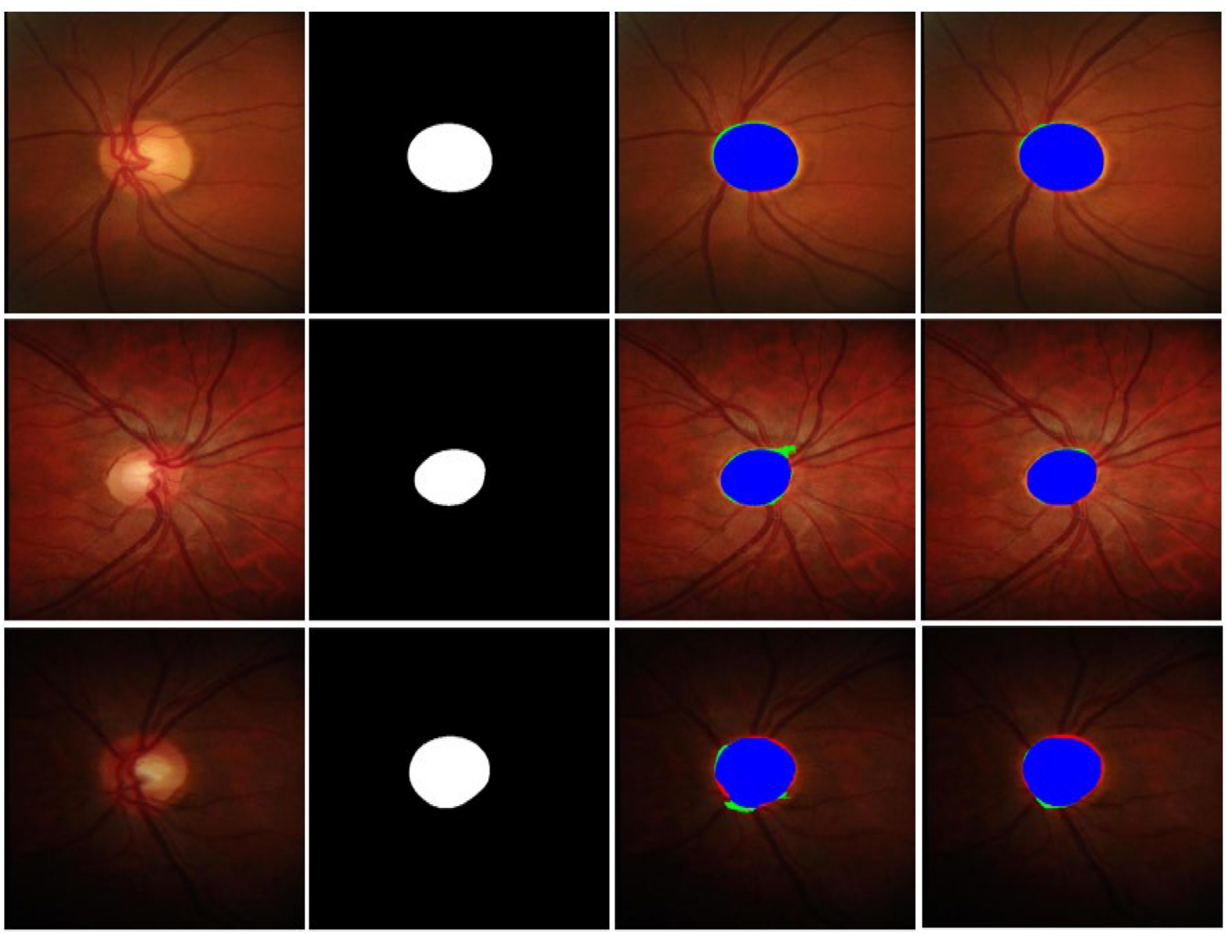
4.3.6. Comparison of ESS-Net and FBSS-Net visual results for the OC and OD segmentation on the Drions-DB dataset (ablation studies)

Drions-DB is quite different from the other three datasets. It has different background effects and illumination variations in the OD region which makes its segmentation interesting. To confirm the effectiveness of the proposed methods we expand the evaluation spectrum and evaluated ESS-Net and FBSS-Net using the Drions-DB. The visual results obtained using both proposed methods are presented in Fig. 13. The shown results indicate a significant performance improvement gained by FBSS-Net compared with ESS-Net. In Fig. 13, good segmentation results are



(a) (b) (c) (d)

Fig. 10. Poor OC and OD segmentation visual results on the REFUGE database, attained by proposed ESS-Net and FBSS-Net. (a) Input image, (b) corresponding ground truth image, and segmented output images acquired by applying (c) ESS-Net and (d) FBSS-Net (tp pixels for the OD and OC are indicated in yellow and blue, respectively. Similarly, red and green indicate fn and fp pixels, respectively).

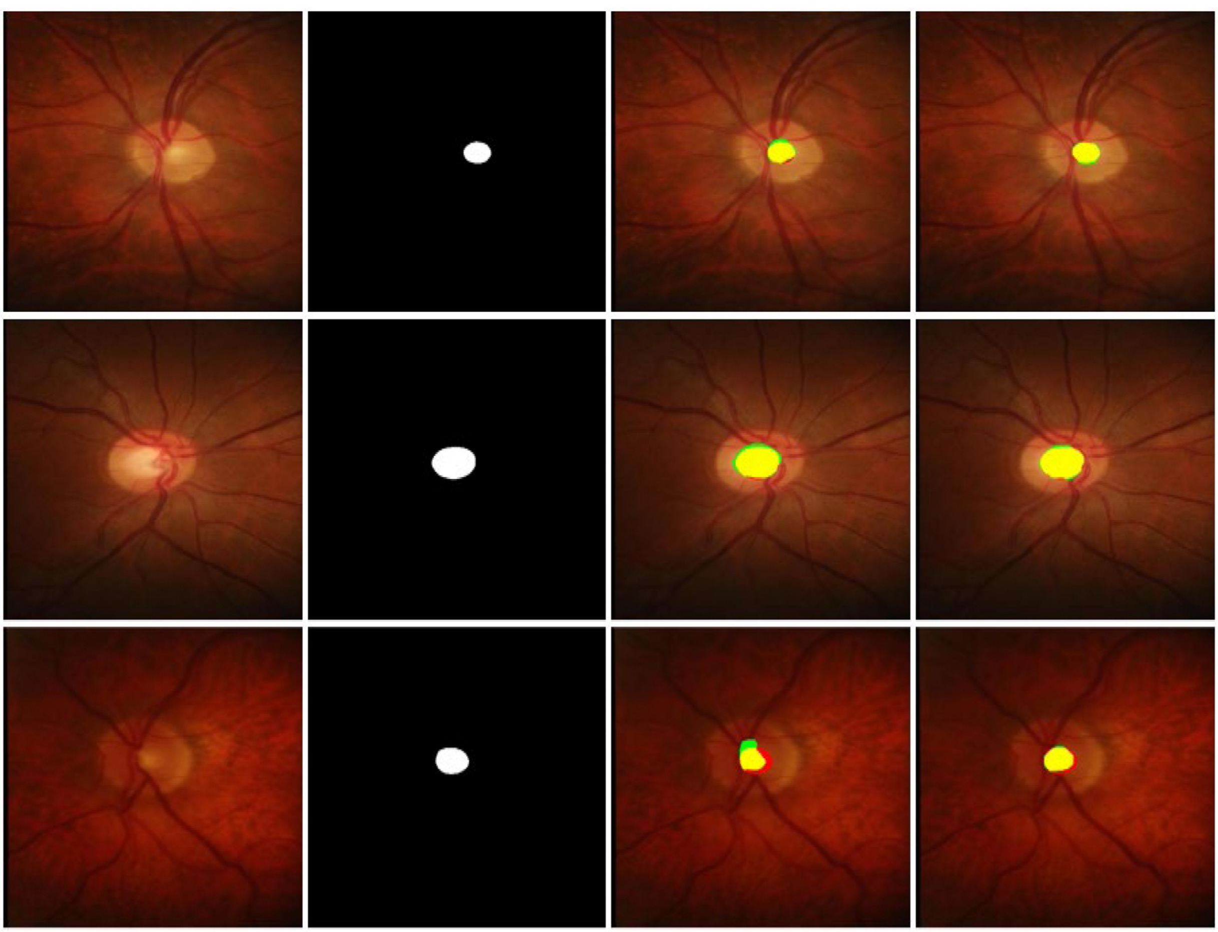


(a) (b) (c) (d)

Fig. 11. Rim-One-r3 database segmentation visual results for the OD, attained using proposed ESS-Net and FBSS-Net: (a) original input image, (b) corresponding ground truth image, (c) segmented output images acquired by applying ESS-Net, and (d) segmented output images obtained by applying FBSS-Net (Rows 1 and 2: good segmentation sample results; Row 3: poor segmentation sample result) (tp and fp pixels are represented with blue and green colors, respectively. fn pixels are shown with red color).

Table 4
Segmentation performance comparison of ESS-Net and FBSS-Net with state-of-the-art methods for REFUGE database. "–" indicates that no result is available (unit: %).

Methods	OC					OD				
	AC	SE	SP	DI	JC	AC	SE	SP	DI	JC
M-Ada [39]	–	–	–	88.25		–	–	–	95.85	–
Denoising [59]	–	–	–	89.4	–	–	–	–	95.3	–
Variational auto-encoder [60]	–	–	–	88.91	–	–	–	–	95.81	–
Patch-based pOSALseg-T [38]	–	–	–	88.2	–	–	–	–	96.0	–
U-Net + VGG16 encoder [61]	–	–	–	–	–	99.8	95.7	–	94.0	89.0
Mask-RCNN [62]	–	–	–	85.4	–	–	–	–	94.7	–
ET-Net [63]	–	–	–	89.1	–	–	–	–	95.2	–
U-Net [32]	–	–	–	85.4	–	–	–	–	93.0	–
Self-attention [43]				85.36					95.09	
Team masker [49]	–	–	–	88.3	–	–	–	–	94.6	–
Cascaded network [42]	–	–	–	88.04	–	–	–	–	93.31	–
Team BUCT [49]	–	–	–	87.2	–	–	–	–	95.2	–
Segtran (eff-B4) [64]	–	–	–	87.2	–	–	–	–	96.1	–
NAS-U^2-Net [65]	–	–	–	89.63		–	–	–	96.14	–
Multi-modal approach [18]	–	–	–	–	79.02	–	–	–	–	92.25
Conditional GAN [66]	–	–	–	–	80.0	–	–	–	–	88.4
SLS-Net [48]	99.9	82.6	99.9	88.0	79.5	99.8	98.6	99.9	96.2	92.8
SLSR-Net [48]	99.9	94.7	99.9	89.5	81.5	99.8	96.9	99.9	96.5	93.3
ESS-Net (Proposed base)	99.92	85.91	99.98	89.20	81.60	99.88	97.98	99.91	96.26	92.91
FBSS-Net (Proposed final)	99.92	91.95	99.96	90.70	83.53	99.91	97.41	99.95	97.08	94.39



(a) (b) (c) (d)

Fig. 12. Rim-One-r3 database segmentation visual results for the OC, attained using proposed ESS-Net and FBSS-Net: (a) original input image, (b) corresponding ground truth image, (c) segmented output images acquired by applying ESS-Net, and (d) segmented output images obtained by applying FBSS-Net (Rows 1 and 2: good segmentation sample results; Row 3: poor segmentation sample result) (tp and fp pixels are represented with yellow and green colors, respectively. fn pixels are shown with red color).

Table 5
Segmentation performance comparison of ESS-Net and FBSS-Net with other methods for Rim-One-r3 database. "-" represents that no result is available (unit: %).

Methods	OC					OD				
	AC	SE	SP	DI	JC	AC	SE	SP	DI	JC
Patch-based pOSALseg-T [38]	–	–	–	85.6	–	–	–	–	96.8	–
DRIU [67]	–	–	–	–	–	–	–	–	95.5	–
U-Net with modification [34]	–	–	–	–	–	–	–	–	95.0	–
RetinaGAN [37]	–	–	–	–	–	–	–	–	95.5	–
Entropy sampling [68]	–	–	–	82.4	–	–	–	–	94.2	–
Auto-encoder [56]	–	–	–	–	–	99.45	87.30	99.81	90.2	88.24
Edge TPU [33]	–	–	–	84.0	–	–	–	–	85.0	–
SLSR-Net [48]	99.6	87.9	99.8	85.8	75.7	99.7	96.8	99.9	96.9	94.0
ESS-Net (Proposed base)	99.59	89.70	99.73	85.63	75.43	99.69	97.48	99.74	96.48	93.23
FBSS-Net (Proposed final)	99.62	89.63	99.76	86.38	76.61	99.74	97.04	99.86	97.02	94.23

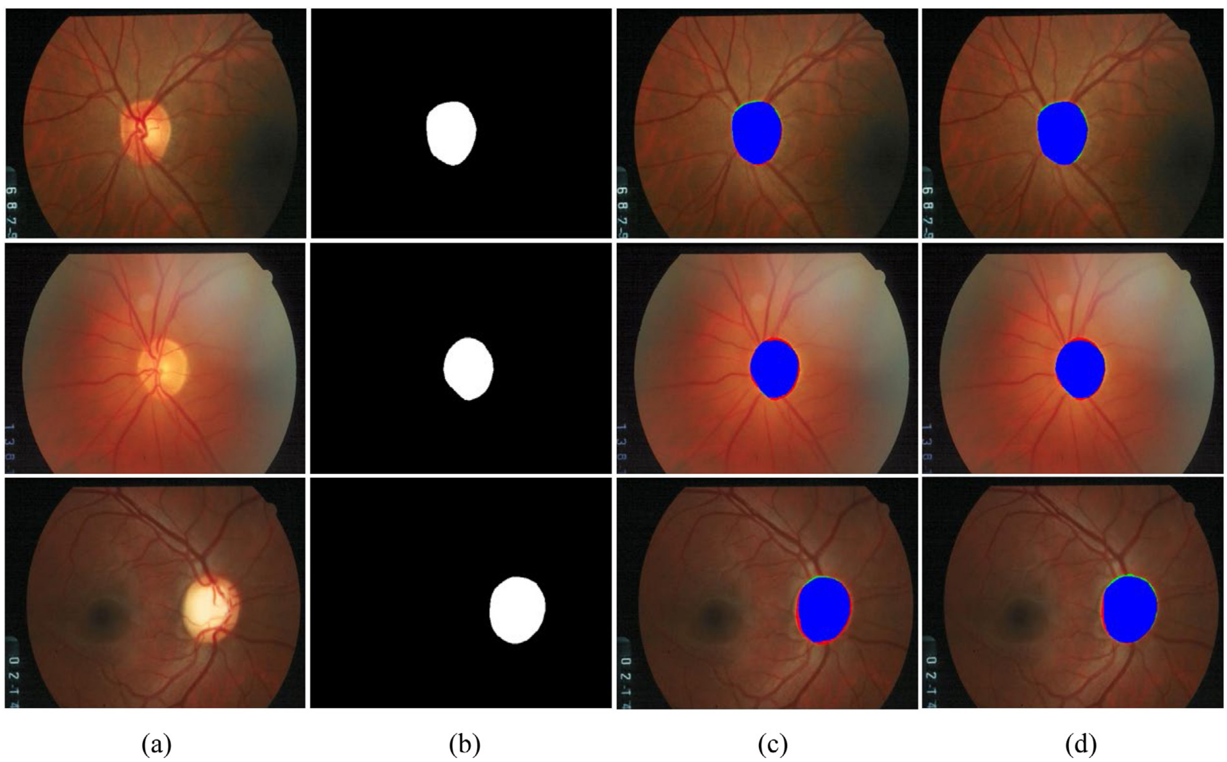


Fig. 13. Segmentation visual results for OD on the Drions-DB dataset attained using proposed ESS-Net and FBSS-Net: (a) original input image, (b) corresponding ground truth image, (c) segmented output images acquired by applying ESS-Net, and (d) segmented output images by applying FBSS-Net (Rows 1 and 2: good segmentation sample results; Row 3: poor segmentation sample result) (tp and fp pixels are represented with blue and green colors, respectively. fn pixels are shown with red color).

presented in rows 1 and 2, whereas a relatively poor segmentation result is shown in row 3. The slightly poor segmentation result was because of the pixel intensity variation near the boundary edges of the OD. However, FBSS-Net delivered better segmentation performance in this case.

4.3.7. Comparing the proposed method on Drions-DB with those of state-of-the-art methods

The segmentation performance achieved by the proposed methods was further compared with those of state-of-the-art methods, and the results are presented in Table 6. The presented results confirmed that FBSS-Net performed better than the state-of-the-art methods. It should be noted that FBSS-Net outperformed the human annotator.

4.3.8. Comparison of ESS-Net and FBSS-Net visual results for the OC and OD segmentation on the Drishti-GS dataset (ablation studies)

The comparative visual results of the OD and OC segmentation are presented in Figs. 14 and 15, respectively. It is worth notable that Drishti-GS images have considerably different contrast values between foreground and background. Similarly, many images from this dataset contain very small-sized OC that is challenging to segment. The visual results confirmed the superior performance of FBSS-Net over ESS-Net. In Figs. 14 and 15, good segmentation results are presented in the first two rows, whereas a relatively poor segmentation result is shown in the last row. The poor segmentation result in Fig. 14 (row 3) was due to an unclear OD boundary. Subsequently, the poor OC segmentation result shown in Fig. 15 (row 3) can be attributed to an indistinctive OC boundary. Nevertheless, FBSS-Net exhibited an improved segmentation performance compared with ESS-Net. In Fig. 16, confusion matrices between FBSS-Net and ESS-Net are presented to demonstrate pixel-wise predictions of the network. Confusion matrices confirm the excellent prediction accuracy delivered by both proposed networks. Nonetheless, FBSS-Net provides overall promising results and significantly outperforms ESS-Net for *tp* and *fn* pixels, especially.

Table 6
Segmentation performance comparison of ESS-Net and FBSS-Net with other methods using the Drions-DB dataset. "-" represents that no result is reported (unit: %).

Methods	AC	SE	SP	DI	JC
DRIU [67]	–	–	–	96.7	–
U-Net [32]	–	–	–	96.8	–
RetinaGAN [37]	–	–	–	96.8	–
[a]The human annotator	–	–	–	96.3	–
SLSR-Net [48]	99.7	95.1	99.9	96.9	94.1
ESS-Net (Proposed base)	99.76	94.45	99.95	96.44	93.17
FBSS-Net (Proposed final)	99.82	97.41	99.91	97.40	94.95

[a]Human annotator results were obtained from [37].

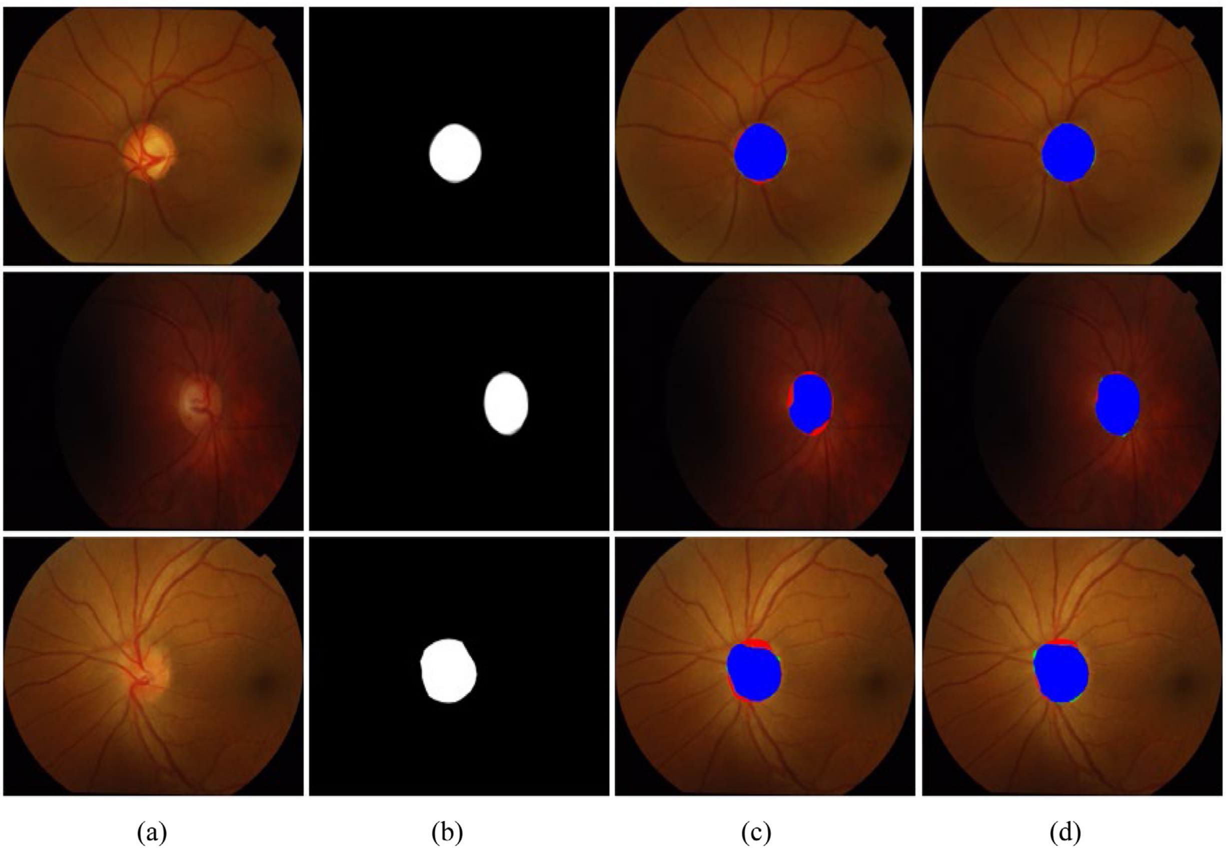


Fig. 14. Segmentation visual results for the OD on the Drishti-GS database, attained by applying ESS-Net and FBSS-Net, (a) original input image, (b) corresponding ground truth image, (c) segmented output images acquired using ESS-Net, and (d) segmented output images obtained using FBSS-Net (Rows 1 and 2: good segmentation sample results; Row 3: poor segmentation sample result) (tp and fp pixels are represented with blue and green colors, respectively. fn pixels are shown with red color).

4.3.9. Comparing the proposed method performance on Drishti-GS database with those of state-of-the-art methods

A comparative analysis between the proposed and state-of-the-art methods is presented in Table 7. It confirms that better segmentation performance was achieved by FBSS-Net compared with state-of-the-art methods. As described earlier, OC segmentation is more challenging because of its indistinctive boundary. Table 7 confirms that FBSS-Net exhibited a significant performance difference for the challenging OC segmentation.

5. Discussion

Glaucoma is a severe retinal disease. Ophthalmologists typically perform manual examinations to screen and diagnose glaucoma. We developed two architectures to provide computer-assisted glaucoma screening through accurate OD and OC segmentations. The proposed architecture provides accurate segmentation with high computational efficiency. Accurate segmented OD and OC images can also aid in other diseases diagnosis.

5.1. Computational efficiency comparison

Computational efficiency is a key measure for the implementation of a developed model. In Table 8, the number of required trainable parameters is presented for a comparative interpretation. The table shows that FBSS-Net exhibited the highest DI with the minimum number of required trainable parameters. FBSS-Net outperformed other methods in segmentation performance and had a superior computational efficiency (parameters: 3.02 M, network size: 4.08 MB).

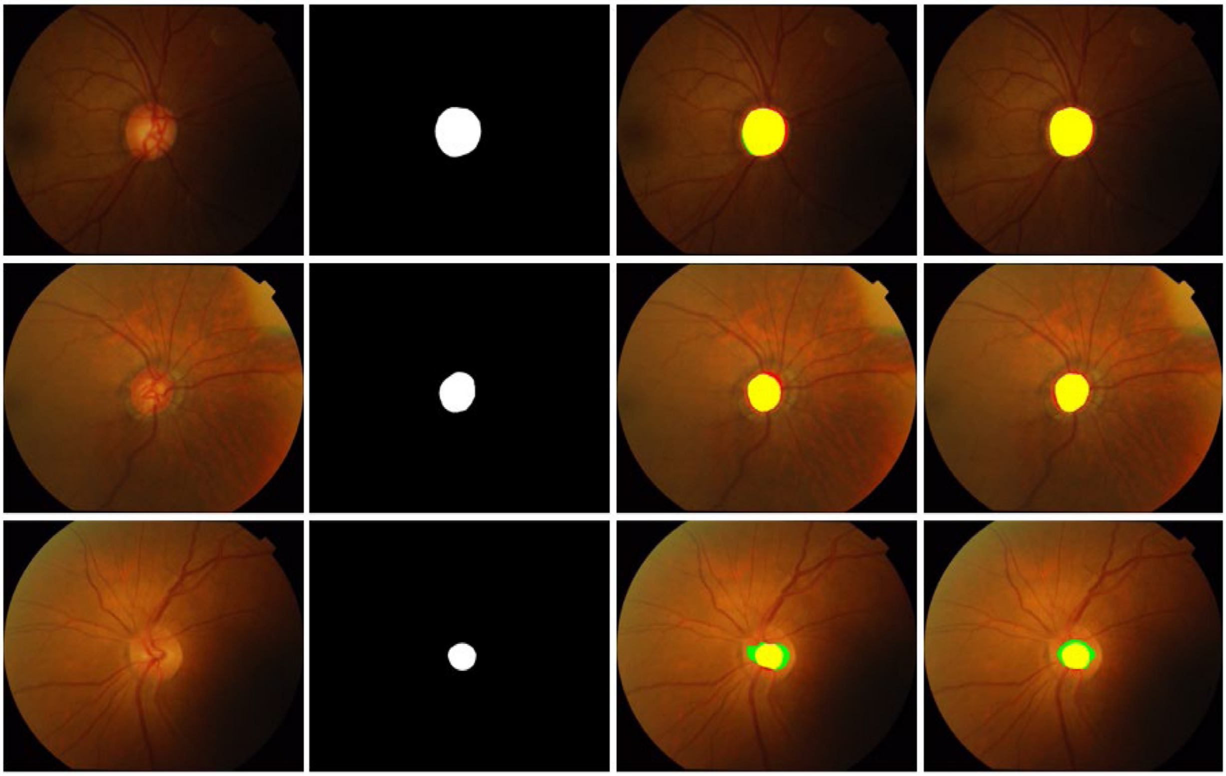


Fig. 15. Segmentation visual results for the OC on the Drishti-GS database, attained by applying ESS-Net and FBSS-Net, (a) original input image, (b) corresponding ground truth image, (c) segmented output images acquired using ESS-Net, and (d) segmented output images obtained using FBSS-Net (Rows 1 and 2: good segmentation sample results; Row 3: poor segmentation sample result) (tp and fp pixels are represented with yellow and green colors, respectively. fn pixels are shown with red color).

(a)

Actual \ Pred	OD	BG
OD	5920750	127591
BG	106261	179441296

Actual \ Pred	OC	BG
OC	3494876	229020
BG	273873	181641596

(b)

Actual \ Pred	OD	BG
OD	5795030	210215
BG	51813	179544068

Actual \ Pred	OC	BG
OC	3458622	262872
BG	267637	181658682

Fig. 16. Confusion matrices for the pixel-wise segmentation of OD and OC with Drishti-GS using (a) FBSS-Net and (b) ESS-Net (Pred represents Prediction).

5.2. Visual interpretation of progressive learning

Gradient-weighted class activation mapping (Grad-CAM) [81] was employed for the visual interpretation of the features considered by FBSS-Net. Grad-CAM highlights the main features considered by a network using heat-activation maps. In Fig. 17, the activation maps for the OD and OC are presented in rows 1 and 2, respectively. Heat activation maps were extracted from the different stages of the proposed network to analyze progressive learning. The Grad-CAM highlights the main features (desired class features) in red. The activation maps in Fig. 17(c)–(f) show the progressive learning of the proposed network for OD and OC features. Fig. 17(f) shows activation maps extracted from the final layer of FBSS-Net, which shows that the proposed network was

Table 7
Comparison of segmentation performance of ESS-Net and FBSS-Net with other methods using Drishti-GS dataset. "–" represents that no result is reported (unit: %).

Methods	OC					OD				
	AC	SE	SP	DI	JC	AC	SE	SP	DI	JC
SU-Net [69]	–	–	–	90.7	–	–	–	–	97.6	–
Multi modal approach [18]	–	–	–	90.29	82.29	–	–	–	96.07	92.43
Patch-based pOSALseg-T [38]	–	–	–	90.1	–	–	–	–	97.4	–
U-Net with VGG16 encoder [61]	–	–	–	–	–	99.79	97.54	–	96.5	93.1
Mod U-Net [40]	–	–	–	88.7	80.4	–	–	–	97.3	94.9
RACE-Net [70]	–	–	–	87.0	–	–	–	–	97.0	–
U-shaped CNN [71]	–	91.5	99.8	89.2	82.3	–	97.9	99.9	97.8	94.9
RetinaGAN [37]	–	–	–	–	–	–	–	–	96.7	–
GL-Net [72]	–	–	–	90.5	–	–	–	–	97.1	–
RFC-Net [35]	97.78	97.82	97.87	90.58	–	97.64	95.78	97.83	97.87	–
FCN [46]	–	–	–	89.7	–	–	–	–	96.7	–
Entropy sampling [68]	–	–	–	87.1	–	–	–	–	97.3	–
Denoising [59]	–	–	–	85.9	–	–	–	–	96.1	–
FC-DenseNet [45]	99.4	–	–	82.8	71.1	99.6	–	–	94.9	90.4
Depth estimation [73]	–	–	–	83.0	–	–	–	–	97.0	–
FCN and adversarial [74]	–	–	–	85.0	75.0	–	–	–	–	–
Edge smoothing approach [75]	–	–	–	81.0	–	–	–	–	95.0	–
Modified U-Net [34]	–	–	–	85.0	–	–	–	–	–	–
Shape regression [76]	–	–	–	85.0	–	–	–	–	95.0	–
Multi-Stage framework [77]	–	–	–	84.0	–	–	–	–	97.0	–
Drishti-GS Challenge [52]	–	–	–	79.0	–	–	–	–	96.0	–
SegNet [78]	99.6	92.0	99.8	88.9	82.4	99.8	98.6	99.9	97.9	96.0
Edge TPU [33]	–	–	–	88.0	–	–	–	–	90.0	–
U-Net [32]	98.1	86.5	98.3	70.2	57.8	99.2	92.8	99.4	91.3	85.8
NAS-U^2-Net [65]	–	–	–	87.69	–	–	–	–	96.95	–
SLS-Net [48]	99.6	93.0	99.8	90.0	82.8	99.8	95.7	99.9	97.4	95.0
SLSR-Net [48]	99.7	94.7	99.8	91.5	85.3	99.8	98.6	99.9	98.0	96.2
ESS-Net (Proposed base)	99.71	92.69	99.85	90.91	84.48	99.85	96.45	99.97	97.74	95.64
FBSS-Net (Proposed final)	99.72	94.0	99.84	91.75	85.61	99.87	97.87	99.93	98.04	96.19

Table 8
Comparing trainable parameters and accuracies of the proposed methods with other methods using REFUGE dataset. (M: million; MB: Megabyte).

Methods	Number of parameters	DI (%)		Network size
		OC	OD	
Team masker [49]	1,224 M	88.3	94.6	–
Mask-RCNN [62]	127 M	85.4	94.7	–
Segtran (eff-B4) [64]	93.1 M	87.2	96.1	–
pOSAL(Xception)* [38]	41.3 M	88.5	95.3	–
U-Net [32]	31.03 M	85.4	93.0	–
U-Net with VGG16 encoder [61]	16.8 M	–	94.0	–
pOSAL (MobileNetV2)[a] [38]	5.8 M	88.5	95.6	–
SLSR-Net [48]	4.67 M	89.5	96.5	16.8 MB
FBSS-Net (proposed)	**3.02 M**	**90.70**	**97.08**	**4.08 MB**

[a]In pOSAL, MobileNetV2 [79] and Xception [80] were employed as backbones.

focused on learning the desired class (OD/OC) features without any biases.

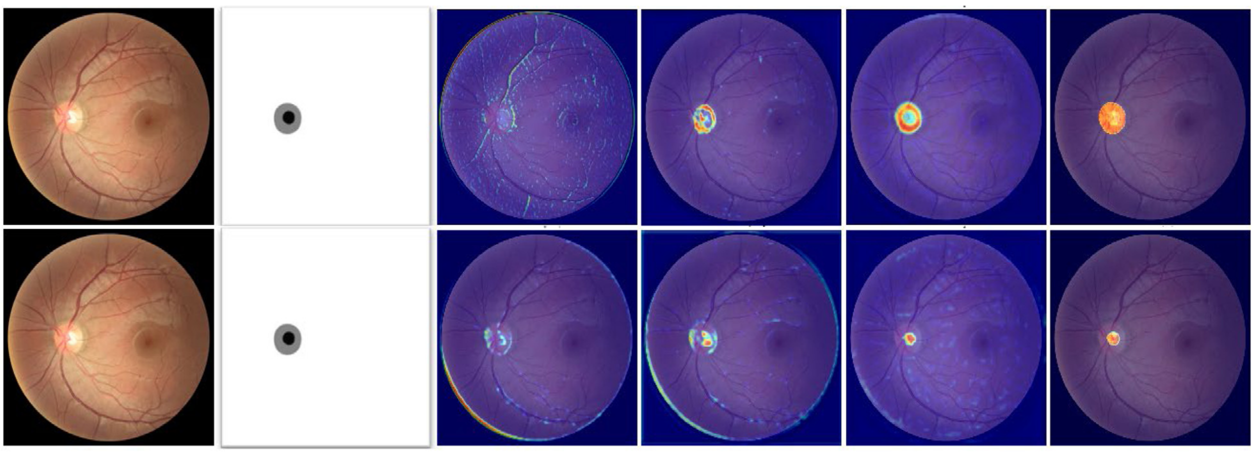
5.3. vCDR-based assistance in glaucoma screening

The vCDR is the ratio of the vertical diameter of the cup to that of the disc. Many studies have reported that the vCDR is one of the key biomarkers for automated diagnosis of glaucoma [38,40,49,82]. In glaucoma patients, the vCDR changes with disease progression, and accurate vCDR computation is based on the accurate segmentation of the OC and OD. Mathematically, the vCDR can be expressed as

$$vCDR = \frac{VD_c}{VD_d} \tag{11}$$

VD_c and VD_d represent the vertical diameters of OC and OD, respectively. A high vCDR value refers to a high glaucoma probability [76]. An illustration of the vCDR computation is shown in Fig. 18(a). CDR_p is the vCDR predicted by FBSS-Net, whereas CDR_g is the vCDR value for the ground truth image shown in Fig. 18(b). CDR_p was computed as 0.84 by FBSS-Net, which is a high vCDR value and was therefore classified as a glaucomatous case. A CDR_g value of 0.835 was sufficiently close to CDR_p, and it exhibited a high segmentation performance with our proposed method.

The segmentation-based classification results using FBSS-Net are provided in Tables 9 and 10 for the Rim-One-r3 and REFUGE datasets, respectively. F1 is the mean of precision and recall as given in [16], and we also included it in evaluation measures for the classification performance evaluation. Glaucoma

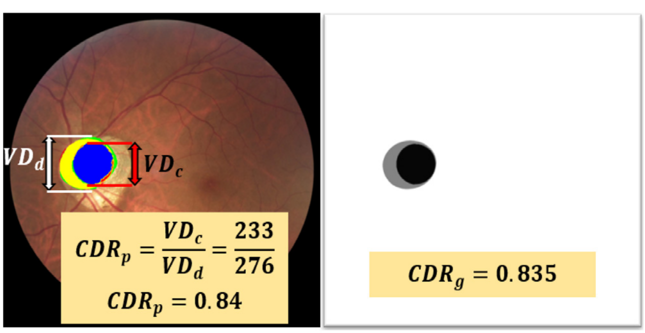


(a) (b) (c) (d) (e) (f)

Fig. 17. FBSS-Net progressive learning on the REFUGE dataset, shown with the aid of heat activation maps extracted from four different stages of the network. Heat activation maps for the OD and OC are presented in rows 1 and 2, respectively. (a) Input image. (b) Ground truth image. Heat activation maps extracted from (c) ReLU-1, (d) ReLU-4, (e) ReLU-8, and (f) ReLU-12 layers of the network, as given in Table 2.

Table 9
Comparing glaucoma screening using non-segmentation (Resent-18 and Resnet-50) and semantic segmentation (FBSS-Net) methods with Rim-One-r3 database (unit: %).

Methods	AC	SP	SE	F1
Classification with Resnet-18	85.0	97.8	42.8	59.6
Classification with Resnet-50	78.3	91.3	35.7	51.3
SLSR-Net [48] (segmentation-based glaucoma classification)	90.0	93.4	78.5	85.3
Proposed FBSS-Net (segmentation-based glaucoma classification)	95.0	98.9	80.0	88.88



(a) (b)

Fig. 18. VCDR-based glaucoma assistance. (a) Original image and (b) ground truth image (VD_c represents the vertical diameter of the OC, whereas VD_d represents the vertical diameter of the OD. CDR_p refers to the predicted vCDR, and CDR_g denotes the vCDR for the corresponding ground-truth image).

screening assistance can also be achieved by using direct classification algorithms. Therefore, we also performed experiments to classify glaucoma and non-glaucoma cases using the well-known Resnet-50 and Resnet-18. For classification, we followed the same data splits and training protocols for a fair comparison with segmentation-based classification. Tables 9 and 10 present a quantitative comparison between the proposed segmentation-based classification (FBSS-Net) and direct classification on the Rim-One-r3 and REFUGE datasets. The presented results indicate that the proposed network had superior classification performance compared to other methods. It should also be noted that FBSS-Net outperformed SLSR-Net for most of the evaluation metrics for segmentation-based classification.

5.4. Assisting the glaucoma screening process (non-vCDR approaches)

Many studies have reported the vCDR as a gold standard for automatic glaucoma screening. In addition to the vCDR computation, a summary of some non-vCDR techniques is presented to broaden the automated glaucoma screening spectrum. The areas of the OC and OD regions can also be employed as biomarker for glaucoma detection [83]. The ratio of the cup area to the disc area indicates the A-CDR [83]. Unlike the vCDR, the A-CDR does not require any specific reference point for its computation, despite the provision of an area ratio, irrespective of any reference bias. Accurate segmentation is required for the accurate computation of the A-CDR. The increased area of the cup itself refers to a high chance of glaucoma [29]. FBSS-Net can provide accurate OD and OC areas through pixel-wise segmentation. The area between the OC and OD is referred as neural rim. The phenomenon associated with shrinking neural rim is called notching [84]. Notching assessments can also assist in glaucoma screening. Subsequently, the ISNT rule also enables the discrimination of glaucomatous and non-glaucomatous cases based on the neural rim width [85]. ISNT rule states that, the neural rim width must be thickest to thinnest for the inferior superior nasal and temporal regions in the order I > S > *N* > T [85]. All cases following the ISNT rule are classified as normal or otherwise as glaucomatous [85]. Fig. 19 shows a sample image from the REFUGE dataset to demonstrate the ISNT rule. In the segmented image, the neural rim was extracted by subtracting the OC from the OD region. Fig. 19(d) presents the zoomed neural rim with corresponding ISNT labels. The neural rim was thinner in the I and S regions and thicker in the *T* region. This case violated the ISNT rule and was therefore classified as a glaucomatous case.

Similarly, the DDLS has a vital role in disease detection and severity presumption [86]. The DDLS is computed using the ratio of the thinnest neural rim width to the disc diameter [86].

Table 10
Comparing glaucoma screening using non-segmentation (Resent-18 and Resnet-50) and semantic segmentation (FBSS-Net) methods with REFUGE database (unit: %).

Methods	AC	SP	SE	F1
Classification with Resnet-18	88.7	91.3	65.0	75.9
Classification with Resnet-50	91.7	94.4	67.5	78.7
SLSR-Net [48] (segmentation-based glaucoma classification)	92.5	95.2	67.5	79.0
Proposed FBSS-Net (segmentation-based glaucoma classification)	95.25	98.33	67.5	80.05

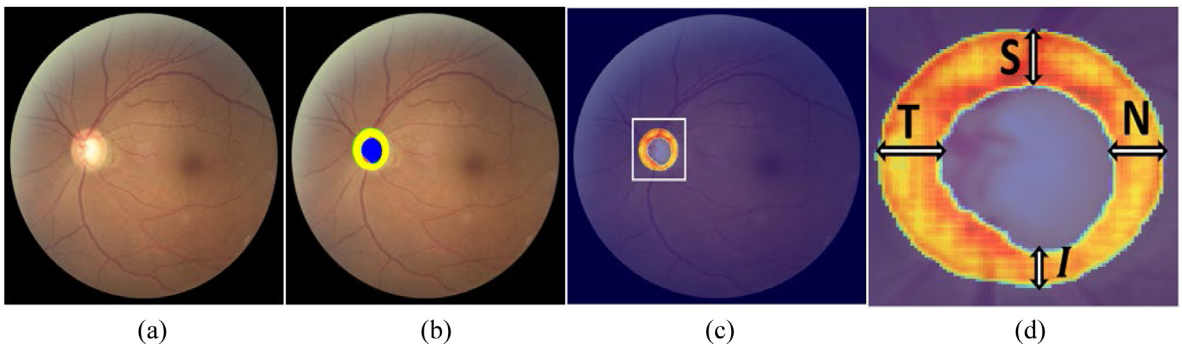


Fig. 19. ISNT rule illustration (a) input image. (b) Segmented image by FBSS-Net. (c) Extracted neural rim, and (d) zoomed neural rim with illustration.

All the presented vCDR or non-vCDR-based glaucoma screening methods are based on pixel-wise segmentation. Therefore, the accurate segmentation of OC and OD is crucial for the accurate computation of the aforementioned schemes.

5.5. OD and OC segmentation for other diseases diagnosis

OD and OC segmentation is not limited to glaucoma detection; instead, it is used as an important tool for the diagnosis of other diseases. Alzheimer's disease (AD) is a common disease based on a neurodegenerative disorder that occurs during the aging process [87]. AD patients exhibit uncomfortable sleep, memory loss, and poor understanding as the main symptoms [87]. As the brain and eye share a common anatomy, retinal analysis can be effective for AD diagnosis. In retinal assessment, the vCDR is the main computational biomarker, and patients with vCDR values of 0.6 or higher are likely to have AD [87]. Similarly, poor cognitive functioning is common among postmenopausal women [88]. The vCDR is considered by medical experts to test for poor cognitive functioning [88]. Hence, accurate OC and OD segmentation assists in glaucoma screening and the diagnosis of many other diseases.

6. Conclusion

The healthcare sector has undergone revolutionary development using computer-assisted biomedical solutions. Glaucoma is a common and critical retinal disease that can cause irreversible loss of vision. OD and OC segmentation is widely accepted for computer-assisted automated glaucoma screening. Two novel segmentation architectures (ESS-Net and FBSS-Net) were developed for the accurate segmentation of the OC and OD. ESS-Net has a shallow architecture with a maximum-depth semantic preservation block, which aids in accurate boundary predictions. FBSS-Net is the final network of this study, and it employs internal and external feature blending to improve the overall segmentation performance. Internal feature blending ensures feature empowerment after intervals, whereas external feature blending enables the network to enhance learning, resulting in improved performance. The proposed networks exhibit superior computational efficiency and require only 3.02 million trainable parameters.

Evaluation of both proposed methods is conducted using four public databases. FBSS-Net delivered excellent segmentation performance with superior computational efficiency. Despite providing excellent results, the proposed methods have certain limitations and boundary conditions. A relatively low segmentation performance of OC because of its indistinctive boundary can be attributed as the common limitation for all methods including the proposed one. In addition, limited annotated medical data availability and the requirement of augmentation can also be considered further limitations. In future work, we intend to develop more effective and efficient models to further enhance the performance, especially for OC with the reduced requirement of trainable parameters. Moreover, we also intend to apply developed networks to other retinal diseases such as diabetic retinopathy and corneal ulcers.

CRediT authorship contribution statement

Adnan Haider: Design methodology, Implementation, Writing of the manuscript. **Muhammad Arsalan:** Comparative analysis. **Chanhum Park:** Literature analysis. **Haseeb Sultan:** Data preprocessing. **Kang Ryoung Park:** Reviewing and editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was supported in part by the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT (MSIT) through the Basic Science Research Program (NRF-2021R1F1A1045587), in part by the NRF funded by the MSIT through the Basic Science Research Program (NRF-2022R1F1A1064291), and in part by the MSIT, Korea, under the ITRC (Information Technology Research Center) support program (IITP-2022-2020-0-01789) supervised by the IITP (Institute for Information & Communications Technology Planning & Evaluation).

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